

Overall Scope: Design Module 1 — Data Acquisition for an IoT-Based Intelligent Electric Vehicle:

You're designing and simulating a **complete data-acquisition system** for an EV, focusing on sustainability, safety, and remote IoT monitoring.

- Plus your **MATLAB/Simulink + Simscape Simulation file** (5 marks).

Purpose: Summarize your entire system visually.

Required elements:

A. Sensor Selection (1 mark):

Choose ≥ 5 sensors relevant to an intelligent EV:

- **Temperature (battery/engine)**
- **Current sensor (Hall-effect)**
- **Voltage divider for battery pack**
- **Speed/RPM sensor (Hall or optical)**
- **Tire-pressure transducer**

For each, include manufacturer + datasheet values (range, sensitivity, output type).

B. Signal Conditioning (2 marks)

- **Amplifier design:** Op-amp type, gain formula, resistor values.
- **Filter:** 1st/2nd order low-pass or Butterworth; specify f_c , R C values.
- Provide circuit diagrams (Simscape > Op-Amp or Simulink).
- Show equations and calculations.

C. Multiplexer (1 mark) (you can use the port

ample-and-Hold & ADC (1 mark)

D. IDE Description

Briefly outline setup: analogRead(), Serial.print(), delay().

Mention analog input range (0–1023), 10-bit resolution.

E. IoT Platform Overview

Describe data upload, visualization, and alert mechanism.

Alignment Checklist for Full Marks:

- ≥ 5 sensors with datasheets
- Amplifier + filter with calculated values
- Multiplexer truth table (s2 s1 s0 → 8 channels)
- Sample-and-hold + ADC design and equations
- Integrated block diagram showing data flow
- Simulation showing realistic signals in Simulink / Simscape

s2	s1	s0	Selected Input / Action	Description / Use Case
0	0	0	Temperature Sensor (Engine Heater)	Reads engine temp; controls heater PWM
0	0	1	Tire Pressure Sensor	Reads tire pressure; triggers warning LED

0	1	0	Water Meter Circuit	Measures flow / level
0	1	1	Portioning System Sensor	Reads the feed portion size or weight
1	0	0	Empty Sensor (0 Value)	Default or calibration
1	0	1	Heater Fault Feedback	Monitors over-temperature / open circuit
1	1	0	Pressure Warning Override	Forces tire warning for system test
1	1	1	Diagnostic Mode	Used for debugging or test signal input

Simulation

Temperature Sensor:

You can use the sine wave in Simulink.

Signal Generator (Sine Wave)

Amplitude = 50 % ΔT around baseline

Bias = 50 % baseline 50°C

Frequency = 0.01 % slow variation, like environmental change

Voltage Sensor (Battery Monitoring):

Signal Generator (Step or Ramp) (Use a **Step** block)

- Initial Value = 4.8
- Final Value = 4.2
- Step Time = 30

Type of Signal	Block	Typical Use	Example Settings
Step	Step	Sudden voltage/temperature change	4.8 → 4.2 V at t = 30 s
Ramp	Ramp	Gradual discharge/heating	Slope = -0.01, Start = 0
Cyclic	Repeating Sequence	Repeated charge/discharge	[0 10 30 60], [4.8 4.8 4.2 4.8]
Periodic	Signal Generator	Sine fluctuations	Amplitude = 1, Freq = 0.1 Hz
Periodic	Sine Wave	Sine fluctuations	Amplitude = 1, Freq = 0.1 Hz

For your Multiport switch, you can use the Control Signal (Repeating sequence)

